MAA HL

Test on Exponents and Logarithms (1)

(without GDC)

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Date: 28 November 2019

Marks:	/40

Name of student: SOLUTIONS

1. [Maximum mark: 6]

Let $\log_3 2 = a$ and $\log_3 7 = b$. Express the following in terms of a and b.

(a) $\log_3 28$ [2]

(b) $\log_3 42$ [2]

(c) $\log_7 \sqrt{2}$ [2]

(a) $\log_3 28 = \log_3 2^2 \cdot 7 = 2\log_3 2 + \log_3 7$

$= \boxed{2a+b}$

(b) $\log_3 42 = \log_3 2\cdot 3\cdot 7 = \log_3 2 + \log_3 3 + \log_3 7$

$= \boxed{a+b+1}$

(c) $\log_7 \sqrt{2} = \dfrac{\log_3 \sqrt{2}}{\log_3 7} = \dfrac{\frac{1}{2}\log_3 2}{\log_3 7} = \boxed{\dfrac{a}{2b}}$

Turn over

2. [Maximum mark: 6]

Solve the logarithmic equation

$$\log_3(x+1) = \log_9(x-1)+1$$

$$\log_3(x+1) = \frac{\log_3(x-1)}{\log_3 9} + 1$$

$$\Leftrightarrow \log_3(x+1) = \frac{\log_3(x-1)}{2} + 1$$

$$\Leftrightarrow 2\log_3(x+1) = \log_3(x-1) + 2$$

$$\Leftrightarrow \log_3 \frac{(x+1)^2}{x-1} = 2$$

$$\Leftrightarrow \frac{(x+1)^2}{x-1} = 3^2$$

$$\Leftrightarrow x^2+2x+1 = 9x-9$$

$$\Leftrightarrow x^2-7x+10 = 0$$

$$\Leftrightarrow \boxed{x=2} \text{ or } \boxed{x=5}$$

3. [Maximum mark: 5]

Solve the system of equations

$$2e^{2x+1} + 5 = y$$

$$\log_7(3y - 2) = 2$$

$\log_7(3y-2) = 2 \Leftrightarrow 3y - 2 = 7^2$

$\Leftrightarrow 3y = 51$

$\Leftrightarrow \boxed{y = 17}$

$2e^{2x+1} + 5 = 17 \Leftrightarrow 2e^{2x+1} = 12$

$\Leftrightarrow e^{2x+1} = 6$

$\Leftrightarrow 2x + 1 = \ln 6$

$\Leftrightarrow \boxed{x = \frac{\ln 6 - 1}{2}}$

Turn over

4. [Maximum mark: 4]

Show that

$$\log_2 7 \cdot \log_7 3 \cdot \log_3 8 = 3$$

$$\log_2 7 \cdot \log_7 3 \cdot \log_3 8 = \frac{\cancel{\ln 7}}{\ln 2} \frac{\cancel{\ln 3}}{\cancel{\ln 7}} \frac{\ln 8}{\cancel{\ln 3}}$$

$$= \frac{\ln 2^3}{\ln 2} = \frac{3\cancel{\ln 2}}{\cancel{\ln 2}} = 3$$

5. [Maximum mark: 6]

Solve the equation

$$\log_{\sqrt{2}}(2\log_4 x \cdot \log_2 x + \log_{\sqrt{2}} x) = 6$$

$2\log_4 x \cdot \log_2 x + \log_{\sqrt{2}} x = \sqrt{2}^6$

$\Leftrightarrow 2\dfrac{\log_2 x}{\cancel{\log_2 4}} \cdot \log_2 x + \dfrac{\log_2 x}{\log_2 \sqrt{2}} = 8$

$(\log_2 x)^2 + 2\log_2 x = 8$

Let $y = \log_2 x$: $y^2 + 2y - 8 = 0$

$\Delta = 4 + 32 = 36$ $\quad y = \dfrac{-2 \pm 6}{2} \Rightarrow y = -4$ or $y = 2$

- $\log_2 x = -4 \Leftrightarrow x = 2^{-4} \Leftrightarrow x = \dfrac{1}{2^4} \Leftrightarrow \boxed{x = \dfrac{1}{16}}$

- $\log_2 x = 2 \Leftrightarrow x = 2^2 \Leftrightarrow \boxed{x = 4}$

Turn over

6. [Maximum mark: 6]

(a) Show that $6xy + 2x - 15y - 5 \equiv (2x-5)(3y+1)$ [1]

(b) **Hence** solve the equation $6x\log_8 x + 2x - 15\log_8 x = 5$ [5]

(a) $(2x-5)(3y+1) = 2x \cdot 3y + 2x - 5 \cdot 3y - 5$

$= 6xy + 2x - 15y - 5$

(b) Let $\log_8 x = y$. Then

$6xy + 2x - 15y = 5 \Leftrightarrow 6xy + 2x - 15y - 5 = 0$

$\Leftrightarrow (2x-5)(3y+1) = 0$

$\Leftrightarrow \boxed{x = \frac{5}{2}}$ or $y = -\frac{1}{3}$

$\Leftrightarrow \log_8 x = -\frac{1}{3}$

$\Leftrightarrow x = 8^{-1/3}$

$\Leftrightarrow \boxed{x = \frac{1}{2}}$

7. [Maximum mark: 7]

The cubic function $f(t) = t^3 - 10t^2 + 7t + 18$ is divisible by $(t^2 - t - 2)$.

(a) Find the three real roots of $f(t)$. [4]

(b) Solve the exponential equation

$$27^x + 7 \cdot (3^x) + 18 = 10 \cdot (9^x)$$ [3]

(a) $t^2 - t - 2 = 0$ $\Delta = 1 + 8 = 9$

$t = \frac{1 \pm 3}{2} \Rightarrow t = 2$ or $t = -1$

Sum of roots: $S = \frac{10}{1} = 10$

Hence $2 + (-1) + x_3 = 10 \Leftrightarrow x_3 = 9$

The three roots are $-1, 2, 9$

(b) Let $t = 3^x$. Then

$t^3 + 7 \cdot t + 18 = 10t^2 \Leftrightarrow t^3 - 10t^2 + 7t + 18 = 0$

$t = -1 \Rightarrow 3^x = -1$ (rejected)

$t = 2 \Rightarrow 3^x = 2 \Rightarrow \boxed{x = \log_3 2}$

$t = 3 \Rightarrow 3^x = 9 \Rightarrow \boxed{x = 2}$